

Collection of Real Analysis Problems

Zeta Zero

For corrections or to suggest a new problem for this collection, please email `category.theory.for.cats@gmail.com`.

1. Let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ ($n \geq 1$) be the sequence of functions with general term

$$f_n(x) = \frac{n^2 x}{x^2 + n^2}.$$

- a) Prove that the sequence (f_n) converges pointwise on $\mathbb{R}$ and determine its pointwise limit f .
 - b) Show that (f_n) does not converge uniformly on $\mathbb{R}$.
 - c) Show that $\int_0^x f(t) dt = \lim_{n \rightarrow \infty} \int_0^x f_n(t) dt$ for every $x \in \mathbb{R}$.
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2. Given a nonempty set $A \subseteq \mathbb{R}$, a function $f : A \rightarrow \mathbb{R}$ is called a *Baire function* if there exists a sequence of functions $f_n : A \rightarrow \mathbb{R}$ ($n \geq 1$), continuous on A , which converges pointwise to f on A .

- a) Prove that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable on $\mathbb{R}$, then its derivative f' is a Baire function.
 - b) Give an example of a Baire function that is not the derivative of any function.
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3. Let $f_n : [a, b] \rightarrow \mathbb{R}$ ($n \geq 1$) be a sequence of functions satisfying:

- (i) all functions f_n are differentiable on $[a, b]$;
 - (ii) the sequence (f_n) converges pointwise on $[a, b]$ to a differentiable function $f : [a, b] \rightarrow \mathbb{R}$;
 - (iii) the sequence (f'_n) converges pointwise on $[a, b]$ to a continuous function $g : [a, b] \rightarrow \mathbb{R}$.
- Prove that $f'(x) = g(x)$ for all $x \in [a, b]$.
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4. Let $g : [0, 1] \rightarrow \mathbb{R}$ be a continuous function with $g(1) = 0$, and let $f_n : [0, 1] \rightarrow \mathbb{R}$ ($n \geq 1$) be the sequence of functions defined by $f_n(x) := x^n g(x)$ for every $n \geq 1$ and every $x \in [0, 1]$. Prove that the sequence (f_n) converges uniformly on $[0, 1]$.

5. Let $A \subseteq \mathbb{R}$ be a nonempty set, and let $f_n : A \rightarrow \mathbb{R}$ ($n \geq 1$) be a sequence of uniformly continuous functions which converges uniformly to $f : A \rightarrow \mathbb{R}$. Prove that f is uniformly continuous.

6. Let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ ($n \geq 1$) be a sequence of functions that converges uniformly on $\mathbb{R}$ to the function $f : \mathbb{R} \rightarrow \mathbb{R}$. Suppose that for each $n \geq 1$ there exists $\ell_n = \lim_{x \rightarrow \infty} f_n(x) \in \mathbb{R}$. Prove that the limits $\lim_{n \rightarrow \infty} \ell_n$ and $\lim_{x \rightarrow \infty} f(x)$ both exist and are equal.

7. Let $f_1 : [a, b] \rightarrow \mathbb{R}$ be a Riemann integrable function, and let $f_n : [a, b] \rightarrow \mathbb{R}$ ($n \geq 1$) be the sequence of functions defined recursively by

$$f_{n+1}(x) = \int_a^x f_n(t) dt \quad \text{for every } x \in [a, b] \text{ and every } n \geq 1.$$

Prove that the sequence (f_n) converges uniformly to the zero function on $[a, b]$.

8. Let $f_n : [0, 1] \rightarrow \mathbb{R}$ ($n \geq 1$) be the sequence defined recursively by $f_1(x) = 0$,

$$f_{n+1}(x) = f_n(x) + \frac{1}{2} (x - f_n^2(x)) \quad \text{for every } x \in [0, 1].$$

Prove that the sequence (f_n) converges uniformly on $[0, 1]$ to the function $f(x) = \sqrt{x}$.

9. Calculate

$$\lim_{n \rightarrow \infty} \int_0^1 \sqrt{\frac{x}{2} + \sqrt{\frac{x}{2} + \cdots + \sqrt{\frac{x}{2} + \sqrt{2}}}} dx$$

with n radicals in the nested expression.

10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Prove that there exists a sequence of polynomial functions $f_n : \mathbb{R} \rightarrow \mathbb{R}$ ($n \geq 1$), uniformly convergent to f , if and only if f is a polynomial function.

11. Let $m \in \mathbb{N}$ and let $f_n : [0, 1] \rightarrow \mathbb{R}$ ($n \geq 1$) be a sequence of polynomial functions of degree at most m , which converge pointwise to the function $f : [0, 1] \rightarrow \mathbb{R}$. Prove that f is a polynomial function of degree at most m , and that the sequence (f_n) converges uniformly to f on $[0, 1]$.

12. Determine the set of convergence of the series of functions

$$\sum_{n \geq 1} \left(\frac{2n^2 + 5}{7n^2 + 3n + 2} \right) \left(\frac{x}{2x + 1} \right)^n, \quad x \in \mathbb{R} \setminus \left\{ -\frac{1}{2} \right\}.$$

13. Define the sequence $x_1 = a \in \mathbb{R}$; $x_{n+1} = x_n^2 + x_n + 1$. Calculate

$$\lim_{n \rightarrow \infty} \frac{(x_{n+1} + 1)^2}{(x_1^2 + 2)(x_2^2 + 2) \cdots (x_n^2 + 2)}.$$

14. Consider the sequence $(x_n)_{n \geq 0}$ defined by $x_0 > 0$; $x_{n+1} = \frac{x_n}{1 + nx_n^2}$. Study the convergence of the sequences $(x_n)_{n \geq 0}$ and $(nx_n)_{n \geq 0}$, and determine their possible limits.

15. Consider the sequence (x_n) of real numbers, defined by $x_1 > 0$; $x_1 + x_1^2 < 1$ and $x_{n+1} = x_n + \frac{x_n^2}{n^2}$ for every $n \geq 1$. Show that the sequences $\left(\frac{1}{x_n} - \frac{1}{n-1} \right)$ and (x_n) are convergent.

16. Let $(a_n)_{n \geq 1}$ be a sequence with strictly positive terms. If

$$a_n \leq a_{2n} + a_{2n+1}, \quad \forall n \in \mathbb{N}^*,$$

study the convergence of the series $\sum_{n=1}^{\infty} a_n$.

17. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$f(x) + f''(x) = -x g(x) f'(x),$$

where $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $g(x) > 0$ for all $x \in \mathbb{R}$. Show that f is bounded.

18. Consider a sequence $(u_n)_{n \geq 1}$ defined by initial terms $u_1, u_2 \in (0, 1)$ and the recursive relation

$$u_{n+2} = \frac{1}{n} u_{n+1}^2 + \frac{n-1}{n} \sqrt{u_n} \quad \text{for all } n \in \mathbb{N}.$$

Determine the limit of the sequence (u_n) as $n \rightarrow \infty$.

19. Let the sequence (u_n) be defined by $u_1 = \frac{1}{2}$ and

$$u_{n+1} = u_n + \frac{u_n^2}{n^2}, \quad \forall n \in \mathbb{N}^*.$$

- a) Prove that $u_n + n^2 \geq \frac{n(n+1)}{\sqrt{2}}$.
b) Prove that (u_n) has a finite limit.
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20. a) Calculate $\int_0^1 x \sin(\pi x^2) dx$.

b) Calculate $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^{n-1} k \int_{k/n}^{(k+1)/n} \sin(\pi x^2) dx$.

21. Let $f : [0, 1] \rightarrow (0, \infty)$ be a continuous function.

a) Prove that for every integer $n \geq 1$ there exists a unique partition $(a_0, a_1, \dots, a_n)$ of $[0, 1]$ such that

$$\int_{a_k}^{a_{k+1}} f(t) dt = \frac{1}{n} \int_0^1 f(t) dt \quad \text{for all } k \in \{0, 1, \dots, n-1\}.$$

b) For each integer $n \geq 1$ define $\bar{a}_n = \frac{a_1 + \dots + a_n}{n}$, where $(a_0, a_1, \dots, a_n)$ is the unique partition from part a). Show that the sequence $(\bar{a}_n)_{n \geq 1}$ is convergent and determine its limit.

22. Let $f : [0, 1] \rightarrow (0, \infty)$ be a continuous function. For each $n \in \mathbb{N}$, $n \geq 2$, consider the partition $(t_0, t_1, \dots, t_n)$ of $[0, 1]$ with the property that

$$\int_{t_0}^{t_1} f(t) dt = \int_{t_1}^{t_2} f(t) dt = \dots = \int_{t_{n-1}}^{t_n} f(t) dt.$$

Determine $\lim_{n \rightarrow \infty} n \left(\frac{1}{f(t_1)} + \frac{1}{f(t_2)} + \dots + \frac{1}{f(t_n)} \right)$.

23. Let $f : [a, b] \rightarrow \mathbb{R}$ be a differentiable function with Riemann integrable derivative. For each $n \in \mathbb{N}$ denote

$$A_n := \frac{b-a}{n} \sum_{k=0}^{n-1} f\left(a + k \frac{b-a}{n}\right), \quad B_n := \frac{b-a}{n} \sum_{k=1}^n f\left(a + k \frac{b-a}{n}\right).$$

Prove that

$$\lim_{n \rightarrow \infty} n \left(A_n - \int_a^b f(x) dx \right) = \frac{b-a}{2} (f(a) - f(b))$$

and

$$\lim_{n \rightarrow \infty} n \left(B_n - \int_a^b f(x) dx \right) = \frac{b-a}{2} (f(b) - f(a)).$$

24. Determine $\lim_{n \rightarrow \infty} n \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} - \ln 2 \right)$.

25. Let $f : [0, 1] \rightarrow \mathbb{R}$ be an increasing, twice differentiable function, with continuous second derivative on $[0, 1]$, and let $(A_n), (B_n)$ be the sequences defined by

$$A_n := \frac{1}{n} \sum_{k=0}^{n-1} f\left(\frac{k}{n}\right), \quad B_n := \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right).$$

The interval $[A_n, B_n]$ is divided into three equal segments. Prove that for n sufficiently large, the number $I := \int_a^b f(x) dx$ belongs to the middle segment.

26. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative, and let

$$s_n = \sum_{k=1}^n f\left(\frac{k}{n}\right) \quad \text{for every } n \in \mathbb{N}.$$

Prove that the sequence $(s_{n+1} - s_n)_{n \geq 1}$ converges to $\int_0^1 f(x) dx$.

27. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be functions with the property that there exist $\alpha \geq 0$ and $p \geq 1$ such that

$$|f(x) - f(x')| \leq \alpha |g(x) - g(x')|^p \quad \text{for all } x, x' \in [a, b].$$

Prove that if g is Riemann integrable, then f is also Riemann integrable.

28. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a Riemann integrable function and let $p \geq 1$. Prove that the limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n \left| f\left(\frac{j-1}{n}\right) - f\left(\frac{j}{n}\right) \right|^p$$

exists, and determine its value.

29. (Riemann's function) Prove that the function $f : [0, 1] \rightarrow \mathbb{R}$, defined by

$$f(x) := \begin{cases} 1 & \text{if } x = 0, \\ 1/q & \text{if } x \in [0, 1] \cap \mathbb{Q}, \ x = p/q, \ p, q \in \mathbb{N}, \ \gcd(p, q) = 1, \\ 0 & \text{if } x \in [0, 1] \setminus \mathbb{Q}, \end{cases}$$

is Riemann integrable. (This function is also known in the literature as Thomae's function, the popcorn function, or the raindrop function.)

30. Prove that if the function $f : [a, b] \rightarrow \mathbb{R}$ has a finite limit at every point of $[a, b]$, then it is bounded.

31. Prove that the set of first-kind discontinuity points of a function $f : I \rightarrow \mathbb{R}$, where $I \subseteq \mathbb{R}$ is an interval, is at most countable.

32. Prove that if the function $f : [a, b] \rightarrow \mathbb{R}$ has a finite limit at every point of $[a, b]$, then it is Riemann integrable.

33. Prove that if $f : [0, 1] \rightarrow \mathbb{R}$ is a Riemann integrable function, then the function $g : [0, 1] \rightarrow \mathbb{R}$, defined by $g(x) = f(x^2)$, is also Riemann integrable.

34. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a Riemann integrable function with the property that $0 < \left| \int_0^1 f(x) dx \right| \leq 1$. Prove that there exist $x_1, x_2 \in [0, 1]$ such that $x_1 \neq x_2$ and

$$\int_{x_1}^{x_2} f(x) dx = (x_1 - x_2)^{2002}.$$

35. Let $f : [0, 1] \rightarrow \mathbb{R}$ be an increasing function and let $n \geq 1$ be an integer. Prove that

$$\int_0^1 f(x) dx \leq (n+1) \int_0^1 x^n f(x) dx.$$

Specify all continuous increasing functions f for which equality holds.

36. Let $f : \left[0, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ be a continuous decreasing function. Prove that

$$\int_{\pi/2-1}^{\pi/2} f(x) dx \leq \int_0^{\pi/2} f(x) \cos x dx \leq \int_0^1 f(x) dx.$$

For which functions f do both inequalities hold with equality?

37. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function and let $M \geq 0$ be such that $|f'(x)| \leq M$ for all $x \in [0, 1]$. Prove that

$$\int_0^1 f^2(x) dx - \left(\int_0^1 f(x) dx \right)^2 \leq \frac{M^2}{12}.$$

38. Let $f_1, \dots, f_n : [0, 1] \rightarrow (0, \infty)$ be continuous functions and let σ be a permutation of $\{1, \dots, n\}$. Prove that

$$\prod_{i=1}^n \int_0^1 \frac{f_i^2(x)}{f_{\sigma(i)}(x)} dx \geq \prod_{i=1}^n \int_0^1 f_i(x) dx.$$

39. Prove that for any continuous function $f : [-1, 1] \rightarrow \mathbb{R}$ the inequality

$$\int_{-1}^1 f^2(x) dx \geq \frac{1}{2} \left(\int_{-1}^1 f(x) dx \right)^2 + \frac{3}{2} \left(\int_{-1}^1 x f(x) dx \right)^2$$

holds. For which functions f does equality hold?

40. a) Given continuous functions $u, v : [0, 1] \rightarrow \mathbb{R}$, prove the Cauchy–Bunyakovsky–Schwarz inequality

$$\left(\int_0^1 u(x)v(x) dx \right)^2 \leq \int_0^1 u^2(x) dx \int_0^1 v^2(x) dx.$$

b) Let $\mathcal{C}$ be the set of all differentiable functions $f : [0, 1] \rightarrow \mathbb{R}$, with continuous derivative f' on $[0, 1]$ and with $f(0) = 0$, $f(1) = 1$. Determine

$$\min_{f \in \mathcal{C}} \int_0^1 \sqrt{1+x^2} (f'(x))^2 dx,$$

as well as all functions $f \in \mathcal{C}$ for which this minimum is attained.

41. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that

$$\int_0^n f(x)f(n-x) dx = \int_0^n f^2(x) dx$$

for every natural number $n \geq 1$. Prove that f is periodic.

42. Let $f : [0, 1] \rightarrow \mathbb{R}$ be an integrable function such that

$$\int_0^1 f(x) dx = \int_0^1 xf(x) dx = 1.$$

Prove that $\int_0^1 f^2(x) dx \geq 4$.

43. Let $\mathcal{M}$ be the set of all continuous functions $f : [0, \pi] \rightarrow \mathbb{R}$, with the property that

$$\int_0^\pi f(x) \sin x dx = \int_0^\pi f(x) \cos x dx = 1.$$

Determine $\min_{f \in \mathcal{M}} \int_0^\pi f^2(x) dx$.

44. Let $p > 1$ be a real number and let $f : [a, b] \rightarrow [0, \infty)$ be a continuous function. Prove that

$$\left(\int_a^b f(x) dx \right)^p \leq (b-a)^{p-1} \int_a^b f(x)^p dx.$$

45. Let $f : [a, b] \rightarrow \mathbb{R}$ be a Riemann integrable function. Prove that the function $\varphi : (0, \infty) \rightarrow \mathbb{R}$, defined by

$$\varphi(r) := \left(\frac{1}{b-a} \int_a^b |f(x)|^r dx \right)^{1/r},$$

is increasing.

46. Let $p > 1$ be a real number. For every continuous function $f : [0, 1] \rightarrow \mathbb{R}$ we denote

$$I(f) := \int_0^1 x^p |f(x)| dx - \int_0^1 x |f(x)|^p dx.$$

Determine the maximum value that $I(f)$ can take.

47. Let $f : [0, 1] \rightarrow \mathbb{R}$ and $g : [0, 1] \rightarrow (0, \infty)$ be continuous functions, with f increasing. Prove that for every $t \in [0, 1]$ the inequality

$$\int_0^t f(x)g(x) dx \int_0^1 g(x) dx \leq \int_0^t g(x) dx \int_0^1 f(x)g(x) dx$$

holds.

48. Let $f : [0, 1] \rightarrow [0, \infty)$ be a decreasing function. Prove that

$$\int_0^1 f(x) dx \int_0^1 x f^2(x) dx \leq \int_0^1 f^2(x) dx \int_0^1 x f(x) dx.$$

49. Prove that if $f : [0, 1] \rightarrow [0, \infty)$ is Riemann integrable, then

$$\int_0^1 f(x) dx \int_0^1 f^2(x) dx \leq \int_0^1 f^3(x) dx.$$

50. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable functions such that $m \leq f(x) \leq M$ for every $x \in [a, b]$ and $\int_a^b g(x) dx = 0$. Prove that

$$\left| \int_a^b f(x)g(x) dx \right| \leq \frac{M - m}{2} \int_a^b |g(x)| dx.$$

51. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative on $[0, 1]$. Prove that if $f(0) = 0$ and $0 < f'(x) \leq 1$ for all $x \in [0, 1]$, then

$$\left(\int_0^1 f(x) dx \right)^2 \geq \int_0^1 f^3(x) dx.$$

Give an example of a function f for which equality holds.

52. a) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative on $[0, 1]$. Prove that if there exists $a \in [0, 1]$ such that $f(a) \leq 0$, then

$$\int_0^1 f(t) dt \leq \int_0^1 |f'(t)| dt.$$

b) Let $g : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative on $[0, 1]$. Prove that for every $x \in [0, 1]$ the inequality

$$\int_0^1 (g(t) - |g'(t)|) dt \leq g(x) \leq \int_0^1 (g(t) + |g'(t)|) dt$$

holds.

53. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative on $[0, 1]$. Prove that if $\int_0^1 f(x) dx = 0$, then for every $\alpha \in (0, 1)$ the inequality

$$\left| \int_0^\alpha f(x) dx \right| \leq \frac{1}{8} \max_{x \in [0, 1]} |f'(x)|$$

holds.

54. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a differentiable function, with the property that there exists $a \in (0, 1)$ such that $\int_{-a}^a f(x) dx = 0$. Denoting $M := \sup_{x \in [-1, 1]} |f'(x)|$, prove that:

- a) $\left| \int_{-1}^1 f(x) dx \right| \leq M(1 - a).$
b) $\int_{-1}^1 |f(x)| dx \geq 2|f(0)| - M.$
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55. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = f(1) = 0$ and $|f'(x)| \leq 1$ for all $x \in [0, 1]$. Prove that

$$\left| \int_0^1 f(x) dx \right| \leq \frac{1}{4}.$$

56. Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function on $[a, b]$, twice differentiable with bounded second derivative on (a, b) , such that $f(a) = f(b) = 0$. Prove that

$$\int_a^b |f(x)| dx \leq M \frac{(b-a)^3}{12}, \quad \text{where } M := \sup\{|f''(x)| : x \in (a, b)\}.$$

57. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a three-times differentiable function with continuous third derivative on $[-1, 1]$, such that $f(-1) = f(0) = f(1) = 0$. Prove that

$$\int_{-1}^1 |f(x)| dx \leq \frac{1}{12} \max_{x \in [-1, 1]} |f'''(x)|.$$

58. (A. Beurling's inequality) Let $f : [a, b] \rightarrow \mathbb{R}$ be twice differentiable with continuous second derivative on $[a, b]$. Prove that if $f(a) = f(b) = 0$, then

$$\int_a^b |f''(x)| dx \geq \frac{4}{b-a} \max_{x \in [a, b]} |f(x)|.$$

59. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative on $[0, 1]$. Prove that if $f(0) = f(1) = -\frac{1}{6}$, then

$$\int_0^1 (f'(x))^2 dx \geq 2 \int_0^1 f(x) dx + \frac{1}{4}.$$

60. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative on $[0, 1]$. Determine f , knowing that $f(1) = -\frac{1}{6}$ and

$$\int_0^1 (f'(x))^2 dx \leq 2 \int_0^1 f(x) dx.$$

61. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative on $[0, 1]$. Prove that if $f\left(\frac{1}{2}\right) = 0$, then

$$\int_0^1 (f'(x))^2 dx \geq 12 \left(\int_0^1 f(x) dx \right)^2.$$

62. Let $f : [a, b] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative on $[a, b]$. Prove that if $f(a) = 0$, then

$$\int_a^b f(x)^2 dx \leq (b-a)^2 \int_a^b f'(x)^2 dx.$$

63. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a twice differentiable function with continuous second derivative on $[-1, 1]$. Prove that if $f(0) = 0$, then

$$\int_{-1}^1 (f''(x))^2 dx \geq 10 \left(\int_{-1}^1 f(x) dx \right)^2.$$

64. Let $n \geq 0$ be an integer and let $f : [-1, 1] \rightarrow \mathbb{R}$ be $2n + 2$ times differentiable with continuous derivative of order $2n + 2$ on $[-1, 1]$. Prove that if $f(0) = f''(0) = \dots = f^{(2n)}(0) = 0$, then

$$\int_{-1}^1 (f^{(2n+2)}(x))^2 dx \geq (2n+2)!^2 (4n+5)^2 \left(\int_{-1}^1 f(x) dx \right)^2.$$

65. Prove that if $f : [0, 1] \rightarrow \mathbb{R}$ is a convex function, then for every $n \in \mathbb{N}$ the inequality

$$\int_0^1 f(x) dx \leq \frac{1}{n+1} \left(f(0) + f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f(1) \right)$$

holds.

66. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a continuous function, twice differentiable on $(-1, 1)$, with the property that

$$2f'(x) + xf''(x) \geq 1 \quad \text{for all } x \in (-1, 1).$$

Prove that $\int_{-1}^1 xf(x) dx \geq \frac{1}{3}$.

67. Show that $(\cos t)^p \leq \cos(pt)$, for every $t \in \left[0, \frac{\pi}{2}\right]$ and $p \in (0, 1)$.

68. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a function of class C^1 , with $f(0) = 0$. Show that

$$\sup_{x \in [0, 1]} |f(x)| \leq \sqrt{\int_0^1 [f'(x)]^2 dx}.$$

69. Let $f : [1, +\infty) \rightarrow \mathbb{R}$ with $f(1) = 1$ and

$$f'(x) = \frac{1}{x^2 + [f(x)]^2}.$$

Show that $\lim_{x \rightarrow \infty} f(x)$ exists and is smaller than $1 + \frac{\pi}{4}$.

70. Let f and $g : [0, 1] \rightarrow (0, +\infty)$ be two continuous functions satisfying the relation

$$\sup_{x \in [0, 1]} f(x) = \sup_{x \in [0, 1]} g(x).$$

Show that there exists $t \in [0, 1]$ with $[f(t)]^2 + 3f(t) = [g(t)]^2 + 3g(t)$.

71. Let $y(h) = 1 - 2\sin^2(2\pi h)$ and $f(y) = \frac{2}{1 + \sqrt{1 - y}}$. Show that

$$f(y(h)) = 2 - 4\sqrt{2\pi} |h| + O(h^2),$$

where $\limsup_{h \rightarrow 0} \frac{O(h^2)}{h^2} < \infty$.

72. Let $f : [0, 1] \rightarrow \mathbb{R}^+$ be a continuous function. Suppose there exist positive constants m and M such that $(f(x) - m)(f(x) - M) \leq 0$ for all $x \in [0, 1]$. Prove that

$$\int_0^1 f(x) dx \int_0^1 \frac{1}{f(x)} dx \leq \frac{(M + m)^2}{4Mm}.$$

73. Let f be a non-negative Riemann integrable function on the interval $[0, 1]$ satisfying $\int_0^1 f(x) dx = 1$. Prove the inequality

$$\int_0^1 \left(x - \int_0^1 u f(u) du \right)^2 f(x) dx \leq \frac{1}{4}.$$

74. Let $f : [1, 2] \rightarrow \mathbb{R}$ be a differentiable function such that its derivative satisfies $|f'(x)| \leq 1$ for all $x \in [1, 2]$. Prove the inequality

$$\left| \int_1^2 f(x) dx - \frac{f(1) + f(2)}{2} \right| \leq \frac{1 - [f(1) - f(2)]^2}{4}.$$

75. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying the Lipschitz condition $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in \mathbb{R}$, where $M > 0$ is a constant. Prove that for any real numbers $a < b$:

- (i) $\left| \frac{1}{b-a} \int_a^b f(x) dx - f\left(\frac{a+b}{2}\right) \right| \leq M \frac{b-a}{4}.$
(ii) $\left| \frac{1}{b-a} \int_a^b f(x) dx - \frac{f(a) + f(b)}{2} \right| \leq M \frac{b-a}{3}.$
-

76. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function. Show that

$$\left| f(1) - \int_0^1 f(x) dx \right| \leq \frac{1}{2} \max_{x \in [0,1]} |f'(x)|.$$

77. Find the smallest constant C such that for every real polynomial $P(x)$ of degree 3 that has a root in the interval $[0, 1]$,

$$\int_0^1 |P(x)| dx \leq C \max_{x \in [0,1]} |P(x)|.$$

78. Let $f : [0, 1] \rightarrow [0, \infty)$ be a non-identically-zero, differentiable function with continuous derivative.

- a) Prove that the sequence $a_n = \int_0^1 \frac{f(x)}{x^n + 1} dx$ ($n \in \mathbb{N}^*$) is convergent.
b) If $a = \lim_{n \rightarrow \infty} a_n$, determine $\lim_{n \rightarrow \infty} \frac{a_n}{a^n}.$
-

79. Let $f : [0, 1] \rightarrow [0, 1]$ be an increasing function and let

$$a_n = \int_0^1 \frac{1 + f^n(x)}{1 + f^{n+1}(x)} dx \quad (n \geq 1).$$

Prove that the sequence $(a_n)_{n \geq 1}$ is convergent and determine its limit.

80. a) Given an integer $n \geq 0$, calculate $\int_0^1 (1-t)^n e^t dt$.

b) Let $k \geq 0$ be an integer and let $(x_n)_{n \geq k}$ be the sequence with general term

$$x_n = \sum_{i=k}^n \binom{i}{k} \left(e - 1 - \frac{1}{1!} - \frac{1}{2!} - \cdots - \frac{1}{i!} \right).$$

Prove that $(x_n)_{n \geq k}$ is convergent and determine its limit.

81. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function and let $(a_n), (b_n)$ be sequences of real numbers with the property that

$$\lim_{n \rightarrow \infty} \int_0^1 |f(x) - a_n x - b_n| dx = 0.$$

Prove that:

a) The sequences (a_n) and (b_n) are convergent.

b) There exist $a, b \in \mathbb{R}$ such that $f(x) = ax + b$ for all $x \in \mathbb{R}$.

82. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function with the property that

$$\int_0^1 f(x)g(x) dx = \int_0^1 f(x) dx \int_0^1 g(x) dx$$

for every continuous, non-differentiable function $g : [0, 1] \rightarrow \mathbb{R}$. Prove that f is constant.

83. Calculate $\lim_{n \rightarrow \infty} \int_0^1 e^{x^n} dx$.

84. a) Prove that $\ln(1+x) \leq x$ for all $x \geq 0$.

b) If $a > 0$, prove that $\lim_{n \rightarrow \infty} n \int_0^1 \frac{x^n}{a+x^n} dx = \ln \frac{a+1}{a}$.

85. Let $f : [0, 1] \rightarrow \mathbb{R}$ be the function $f(x) = (x^2 + 1)e^x$. Calculate

$$\lim_{n \rightarrow \infty} n \int_0^1 \left(f\left(\frac{x}{2^n}\right) - 1 \right) dx.$$

86. For each $n \in \mathbb{N}^*$ consider the function $f_n : [0, n] \rightarrow \mathbb{R}$, defined by $f_n(x) = \arctan([x])$, where $[x]$ denotes the integer part of x . Show that f_n is Riemann integrable and determine

$$\lim_{n \rightarrow \infty} \frac{1}{n} \int_0^n f_n(x) dx.$$

87. Consider the continuous functions $f : [0, \infty) \rightarrow \mathbb{R}$ and $g : [0, 1] \rightarrow \mathbb{R}$. If the limit $\lim_{x \rightarrow \infty} f(x) = L \in \mathbb{R}$ exists, prove that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \int_0^n f(x) g\left(\frac{x}{n}\right) dx = L \int_0^1 g(x) dx.$$

88. For each $\alpha \in (0, 1]$ denote

$$I_n(\alpha) = \int_0^\alpha \ln(1 + x + x^2 + \cdots + x^{n-1}) dx, \quad n \geq 2.$$

Calculate:

- a) $\lim_{n \rightarrow \infty} I_n(\alpha)$ for $\alpha \in (0, 1)$;
 b) $\lim_{n \rightarrow \infty} I_n(1)$.
-

89. Let $\alpha > 0$ and let $a, b \in \mathbb{R}$ be such that $a < b$. Determine

$$\lim_{n \rightarrow \infty} \int_a^b \sqrt[n]{(x-a)^n + (b-x)^n + \alpha((x-a)(b-x))^{n/2}} dx.$$

90. Determine $\lim_{n \rightarrow \infty} n \int_{-1}^0 (x + e^x)^n dx$.

91. Determine $\lim_{n \rightarrow \infty} n \int_0^1 (\cos x - \sin x)^n dx$.

92. Let $\{\epsilon_n\}_{n=1}^\infty$ be a sequence of positive real numbers such that $\lim_{n \rightarrow \infty} \epsilon_n = 0$. Evaluate the limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln\left(\frac{k}{n} + \epsilon_n\right).$$

93. Let α, β be reals such that $2\beta \geq \alpha$ and suppose that they are not simultaneously negative. Let

$$F(\alpha, \beta) = \lim_{n \rightarrow \infty} n^\alpha \int_0^{n^{-\beta}} x^{x+1} dx.$$

Prove that $F(\alpha, \beta)$ exists and is finite. Hence find the value of $F(\alpha, \beta)$.

94. Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be such that

$$\int_x^{x+y} \psi(t) dt = \int_{x-y}^x \psi(t) dt.$$

Find all possible $\psi(x)$.

95. (Wirtinger's inequalities)

a) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative on $[0, 1]$, with the property that $\int_0^1 f(x) dx = 0$. Prove that

$$\int_0^1 f(x)^2 dx \leq \frac{1}{2} \int_0^1 f'(x)^2 dx.$$

b) Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable and satisfy $f(0) = 0$. Prove that

$$\int_0^1 [f'(x)]^2 dx \geq 3 \left(\int_0^1 f(x) dx \right)^2.$$

c) Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable and satisfy $f(0) = f(1) = 0$. Prove that

$$\int_0^1 [f'(x)]^2 dx \geq 12 \left(\int_0^1 f(x) dx \right)^2.$$

d) Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable and satisfy $f(0) = f(1) = 0$. Prove that

$$\int_0^1 [f'(x)]^2 dx \geq \pi^2 \int_0^1 [f(x)]^2 dx.$$

e) Let $f : [0, 1] \rightarrow \mathbb{R}$ be twice continuously differentiable and satisfy $f(0) = f'(0) = 0$. Prove that

$$\int_0^1 [f''(x)]^2 dx \geq 20 \left(\int_0^1 f(x) dx \right)^2.$$

f) Let $f : [0, 1] \rightarrow \mathbb{R}$ be twice continuously differentiable and satisfy $f(0) = f(1) = 0$, $f'(0) = f'(1) = 0$. Prove that

$$\int_0^1 [f''(x)]^2 dx \geq 720 \left(\int_0^1 f(x) dx \right)^2.$$

g) Let $f : [0, 1] \rightarrow \mathbb{R}$ be twice continuously differentiable and satisfy $f(0) = f'(0) = 0$. Prove that

$$\int_0^1 [f''(x)]^2 dx \geq \frac{45}{2} \left(\int_0^1 x f(x) dx \right)^2.$$

h) Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable and satisfy $f(0) = f(1) = 0$. Find the smallest constant C such that

$$\max_{x \in [0, 1]} |f(x)| \leq C \sqrt{\int_0^1 [f'(t)]^2 dt}.$$

i) Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable with $f(0) = f(1) = 0$. Find the best constant C such that

$$|f(c)|^2 \leq C \int_0^1 [f'(x)]^2 dx, \quad \forall c \in [0, 1].$$

j) Let $f : [0, 1] \rightarrow \mathbb{R}$ be twice continuously differentiable and satisfy $f(0) = f(1) = 0$. Find the best constant C such that

$$\int_0^1 |f''(x)| dx \geq C \max_{x \in [0, 1]} |f(x)|.$$

k) Let $f : [0, 1] \rightarrow \mathbb{R}$ be twice continuously differentiable, $f(0) = 0$, $f'(0) = 0$. Find the best constant C such that

$$f(1)^2 \leq C \int_0^1 [f''(x)]^2 dx.$$

l) Let $M = \{f \mid f \in C^2[0, 1], f(0) = f(1) = 0, f'(0) = 1\}$. Determine

$$\min_{f \in M} \int_0^1 (f''(x))^2 dx$$

and the functions for which the minimum is achieved.

m) Let $\mathcal{F}$ be the set of continuously differentiable functions $f : [0, 1] \rightarrow \mathbb{R}$ such that $\int_0^1 f(x) dx = 0$, $f \not\equiv 0$. Find the largest constant K such that

$$\int_0^1 [f'(x)]^2 dx \geq K \left(\int_0^1 x^2 f(x) dx \right)^2,$$

and determine the case of equality.

96. Let $M > 0$ be a real number and let $f : [a, b] \rightarrow \mathbb{R}$ be a twice differentiable function with continuous second derivative on $[a, b]$, with the property that $|f''(x)| \leq M$ for all $x \in [a, b]$. Prove that

$$\left| \frac{1}{b-a} \int_a^b f(x) dx - f\left(\frac{a+b}{2}\right) \right| \leq M \frac{(b-a)^2}{24}.$$

97. Let $a \in [0, 1]$. Determine all closed intervals $I \subset [0, 1]$ with the property that for every differentiable convex function $f : [0, 1] \rightarrow \mathbb{R}$ the inequality

$$\int_0^1 \max\{f(a), f(x)\} dx \geq \max_{x \in I} f(x)$$

holds.

98. Let $f_0 : [0, 1] \rightarrow \mathbb{R}$ be a continuous function, and let f_{n+1} be the antiderivative of f_n on $[0, 1]$ that vanishes at the origin, for $n \in \mathbb{N}$. If $f_{1982}(1) = \frac{1}{1983!}$, prove that there exists $x_0 \in [0, 1]$ such that $f_0(x_0) = x_0$.

99. Let $f : [a, b] \rightarrow \mathbb{R}$ be an integrable function. Prove that:

1) for each $\alpha \in (0, 1)$ there exists at least one $c \in (a, b)$ such that $\alpha \int_a^b f(x) dx = \int_a^c f(x) dx$;

2) if f is continuous on (a, b) and $\int_a^b f(x) dx \neq 0$, then for every $n \in \mathbb{N}^*$ there exist n distinct points $x_1, x_2, \dots, x_n$ in the interval (a, b) such that

$$\int_a^b f(x) dx = \frac{n(b-a)}{\frac{1}{f(x_1)} + \frac{1}{f(x_2)} + \dots + \frac{1}{f(x_n)}}.$$

100. Prove that

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{1}{m(m+1) \cdots (m+n)} = \int_{0+0}^1 \frac{e^x - 1}{x} dx.$$

101. (Generalization of the preceding problem) Prove that for every $m \in \mathbb{N}$ the equality

$$\sum_{n_m=1}^{\infty} \cdots \sum_{n_1=1}^{\infty} \sum_{n_0=1}^{\infty} \frac{1}{n_0(n_0 + n_1) \cdots (n_0 + n_1 + \cdots + n_m)}$$

$$= (-1)^{m-1} \left(\int_{0+0}^1 \frac{e^x - 1}{x} dx + \sum_{j=1}^{m-1} \frac{1}{j} \left(1 - e \sum_{i=0}^{j-1} \frac{(-1)^i}{i!} \right) \right)$$

holds.

102. Determine the continuous decreasing functions $f : [0, 1] \rightarrow [0, \infty)$ that simultaneously satisfy the conditions

$$\int_0^1 f(x) dx \leq 1 \quad \text{and} \quad \int_0^1 f(x^3) dx \leq 3 \left(\int_0^1 x f(x) dx \right)^2.$$

103. Define $g_0 = 1$ and $g_{n+1}(x) = 1 + \int_0^x g_n(t - t^2) dt$ for $x \in [0, 1]$.

- a) Show that $0 \leq g_n(x) - g_{n-1}(x) \leq \frac{x^n}{n!}$.
 - b) Deduce the pointwise convergence of (g_n) .
 - c) Prove that the convergence is uniform on $[0, 1]$.
-

104. Let $f \in C[a, b]$ be such that $\int_a^b x^n f(x) dx = 0$ for every $n \in \mathbb{N}$. Show that f is identically zero.

105. Let $f \in C^1[a, b]$ with $f'(a) \neq 0$. For any $x \in (a, b]$, let $\theta(x) \in [a, x]$ be such that

$$\int_a^x f(t) dt = (x - a)f(\theta(x)).$$

Calculate $\lim_{x \rightarrow a} \frac{\theta(x) - a}{x - a}$.

106. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that

$$\int_0^1 f(x) dx = \int_0^1 x f(x) dx = \cdots = \int_0^1 x^{n-1} f(x) dx = 0 \quad \text{and} \quad \int_0^1 x^n f(x) dx = 1.$$

Show that there exists $a \in [0, 1]$ such that $|f(a)| \geq 2^{n(n+1)}$.

107. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function (i.e., $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$) that is integrable on every compact interval of $\mathbb{R}$. Show that there exists $a \in \mathbb{R}$ such that $f(x) = ax$ for all $x \in \mathbb{R}$.

108. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Show that

$$\int_0^1 \int_x^1 f(t) dt dx = \int_0^1 xf(x) dx.$$

109. Let $f_1, f_2, f_3, f_4 \in \mathbb{R}[x]$. Show that

$$F(x) = \int_1^x f_1(t)f_2(t) dt \int_1^x f_3(t)f_4(t) dt - \int_1^x f_1(t)f_4(t) dt \int_1^x f_2(t)f_3(t) dt$$

is a polynomial divisible by $(x - 1)^4$.

110. Let $f : [0, \pi] \rightarrow \mathbb{R}$ be a continuous function such that

$$\int_0^\pi f(x) \cos x dx = \int_0^\pi f(x) \sin x dx = 0.$$

Show that the equation $f(x) = 0$ has at least two roots in $(0, \pi)$.

111. Let $f : [0, 1] \rightarrow (0, \infty)$ be an increasing function. Prove that

$$\int_0^1 xf^2(x) dx \int_0^1 f(x) dx \geq \int_0^1 f^2(x) dx \int_0^1 xf(x) dx.$$

112. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a twice differentiable function such that $|f(x)| \leq 1$ and $|f''(x)| \leq 1$ for all $x \in [-1, 1]$. Show that $|f'(x)| \leq 2$ for all $x \in [-1, 1]$.

113. Determine the maximum of the expression

$$\int_0^1 x^2 f(x) dx - \int_0^1 x(f(x))^2 dx$$

for $f \in C[0, 1]$.

114. Let $I \subseteq \mathbb{R}$ be an open interval with $0 \in I$ and $f : I \rightarrow \mathbb{R}$ a continuous function with $f(0) = 0$, right-differentiable at 0.

a) Show that $\lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(\frac{k}{n^2}\right) = \frac{1}{2} f'_d(0)$.

b) Special case: calculate $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\sqrt{1 + \frac{k}{n^2}} - 1 \right)$.

115. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function. Assume there exists $\lim_{x \rightarrow \infty} (f(x) + f'(x)) = l \in \mathbb{R}$. Show that

$$\lim_{x \rightarrow \infty} f(x) = l \quad \text{and} \quad \lim_{x \rightarrow \infty} f'(x) = 0.$$

116. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice continuously differentiable function. If $f(x+y)f(x-y) \leq f^2(x)$ for all $x, y \in \mathbb{R}$, show that $f(x)f''(x) \leq (f'(x))^2$ for all $x \in \mathbb{R}$.

117. Let $I \subseteq \mathbb{R}$ be an interval and $f : I \rightarrow \mathbb{R}$ an arbitrary function. Show that f has at most a countable set of strict extrema.

118. Suppose the differentiable functions $a, b, f, g : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(x) \geq 0$, $f'(x) \geq 0$, $g(x) > 0$, $g'(x) > 0$ for all $x \in \mathbb{R}$. Let $\lim_{x \rightarrow \infty} a(x) = A > 0$, $\lim_{x \rightarrow \infty} b(x) = B > 0$, $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = \infty$, and assume

$$\lim_{x \rightarrow \infty} \left(\frac{f'(x)}{g'(x)} + a(x) \frac{f(x)}{g(x)} - b(x) \right) = 0.$$

Show that $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{B}{A+1}$.

119. Let $k > 1$ be a real number. Calculate the limits:

a) $L = \lim_{n \rightarrow \infty} \int_0^1 \sqrt[n]{kx + k - 1} dx$;

b) $\lim_{n \rightarrow \infty} n \left(L - \int_0^1 \sqrt[n]{kx + k - 1} dx \right)$.

120. Let $f : [0, 1] \rightarrow \mathbb{R}$ be the function $f(x) = (x^2 + 1)e^x$. Calculate

$$\lim_{n \rightarrow \infty} \int_0^1 \left(f\left(\frac{x}{n}\right) \right)^n dx.$$

121. Determine

$$\lim_{n \rightarrow \infty} \left(\int_0^1 e^{x^2/n} dx \right)^n.$$

122. Assume $f(x) \geq 0$ and $\int_0^\infty f^2(x) dx < \infty$. Define $F(x) = \int_0^x f(t) dt$. Prove that

$$\int_0^\infty \left(\frac{F(x)}{x} \right)^2 dx \leq 4 \int_0^\infty f^2(x) dx.$$

123. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuously differentiable, satisfying $f(0) = 0$, $f(1) = 1$, and $f'(x) > 0$. Define

$$L = \int_0^1 \sqrt{1 + [f'(x)]^2} dx.$$

Prove that

$$\int_0^1 x \sqrt{1 + [f'(x)]^2} dx \leq \frac{L}{2} + \frac{1}{4}.$$

124. (Gronwall's inequality) Let $u(t)$ be a nonnegative continuous function on $[a, b]$. Assume there exist constants $C \geq 0$, $K \geq 0$ such that

$$u(t) \leq C + K \int_a^t u(s) ds, \quad \forall t \in [a, b].$$

Prove that $u(t) \leq Ce^{K(t-a)}$.

125. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Prove that there exists $c \in (a, b)$ such that

$$f(c) \left(\int_c^b f(x) dx - \int_a^c f(x) dx \right) = 0.$$

126. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous and positive. Prove that there exists $c \in (0, 1)$ such that

$$\int_0^c f(x) dx = (1 - c)f(c).$$

127. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous and satisfy $\int_0^1 f(x) dx = 0$. Prove that there exists $c \in (0, 1)$ such that

$$f(c) = \int_0^c f(x) dx.$$

128. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous and assume $f(1) = 0$. Prove that there exists at least one $c \in (0, 1)$ such that

$$f(c) = \int_0^c f(x) dx.$$

129. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous. Prove that for every positive integer n , there exists $c \in (0, 1)$ such that

$$\int_0^c x^{n-1} f(x) dx = \frac{c^n}{n} f(c).$$

130. Let $f : [0, \frac{\pi}{2}] \rightarrow \mathbb{R}$ be continuous and satisfy

$$\int_0^{\pi/2} f(x) \sin x dx = \int_0^{\pi/2} f(x) \cos x dx.$$

Prove that there exists $c \in (0, \frac{\pi}{2})$ such that

$$f(c) = \left(\int_0^c f(x) dx \right) (\sin c + \cos c).$$

131. Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable and $k \neq 0$. Assume $f(0) = 0$ and

$$f(1) = k \int_0^1 e^{k(1-x)} f(x) dx.$$

Prove that there exists $c \in (0, 1)$ such that $f'(c) = kf(c)$.

132. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and assume $\int_a^b f(x) dx \neq 0$. Prove that there exists $c \in (a, b)$ such that

$$f(c) \left[\frac{1}{\int_a^c f(x) dx} - \frac{1}{\int_c^b f(x) dx} \right] = \frac{1}{c-a} - \frac{1}{b-c}.$$

133. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous. Prove that there exists $c \in (0, 1)$ such that

$$2c^2 f(c) \left(\int_0^c f(x) dx - \int_c^1 f(x) dx \right) = \int_0^c f(x) dx \int_c^1 f(x) dx.$$

134. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuously differentiable, $f(0) = 0$, $f(1) = 1$, and $f'(x) + f(x) > 0$. Find the minimum value of

$$I = \int_0^1 \frac{1}{f'(x) + f(x)} dx.$$

135. Under the same hypotheses as the preceding problem, find the minimum value of

$$J = \int_0^1 \frac{1}{[f'(x) + f(x)]^2} dx.$$

136. Let f be continuous on $[0, 1]$ and satisfy $1 \leq f(x) \leq 3$. Prove that

$$\int_0^1 f(x) dx \int_0^1 \frac{1}{f(x)} dx \leq \frac{4}{3}.$$

137. (Polya–Szegő inequality) Let f, g be positive integrable functions on $[0, 1]$ and assume

$$0 < m \leq \frac{f(x)}{g(x)} \leq M.$$

Prove that

$$\int_0^1 f^2(x) dx \int_0^1 g^2(x) dx \leq \frac{(m+M)^2}{4mM} \left(\int_0^1 f(x)g(x) dx \right)^2.$$

138. Let f be continuous on $[0, 1]$ and satisfy $1 \leq f(x) \leq 3$. Find the maximum value of

$$P = \int_0^1 f(x) dx + \int_0^1 \frac{1}{f(x)} dx - \int_0^1 f(x) dx \int_0^1 \frac{1}{f(x)} dx.$$

139. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuously differentiable and satisfy $1 \leq f(x) \leq 3$. Prove that

$$\int_0^1 f(x) dx \int_0^1 \frac{1}{f(x)} dx \leq \min \left\{ \frac{4}{3}, 1 + \frac{1}{\pi^2} \int_0^1 \left(\frac{f'(x)}{f(x)} \right)^2 dx \right\}.$$

140. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be nonconstant, differentiable functions satisfying

$$f(x+y) = f(x)f(y) - g(x)g(y), \quad g(x+y) = f(x)g(y) + g(x)f(y),$$

for all $x, y \in \mathbb{R}$, and $f'(0) = 0$. Prove that $(f(x))^2 + (g(x))^2 = 1$ for all $x \in \mathbb{R}$.

141. Let f be a real function on the real line with a continuous third derivative. Prove that there exists a point $a \in \mathbb{R}$ such that

$$f(a) \cdot f'(a) \cdot f''(a) \cdot f'''(a) \geq 0.$$

142. Suppose that $f : [0, 1] \rightarrow \mathbb{R}$ has a continuous derivative and that $\int_0^1 f(x) dx = 0$. Prove that for every $\alpha \in (0, 1)$,

$$\left| \int_0^\alpha f(x) dx \right| \leq \frac{1}{8} \max_{0 \leq x \leq 1} |f'(x)|.$$

143. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function satisfying $xf(y) + yf(x) \leq 1$ for every $x, y \in [0, 1]$.

a) Show that $\int_0^1 f(x) dx \leq \frac{\pi}{4}$.

b) Find such a function for which equality occurs.

144. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. A point x is called a *shadow point* if there exists a point $y \in \mathbb{R}$ with $y > x$ such that $f(y) > f(x)$. Let $a < b$ be real numbers and suppose that all the points of the open interval $I = (a, b)$ are shadow points, while a and b are not shadow points. Prove that:

a) $f(x) \leq f(b)$ for all $a < x < b$;

b) $f(a) = f(b)$.

145. Let $(a_n) \subset \left[\frac{1}{2}, 1\right]$. Define the sequence $x_0 = 0$, and

$$x_{n+1} = \frac{a_{n+1} + x_n}{1 + a_{n+1}x_n}.$$

Is this sequence convergent? If yes, find its limit.

146. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $|f(x) - f(y)| \leq |x - y|$ for all $x, y \in \mathbb{R}$. Prove that if for any real x the sequence $x, f(x), f(f(x)), \dots$ is an arithmetic progression, then there is $a \in \mathbb{R}$ such that $f(x) = x + a$ for all $x \in \mathbb{R}$.

147. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a continuous function having finite derivative at 0, and let

$$I(h) = \int_{-h}^h f(x) dx, \quad h \in [0, 1].$$

Prove that:

a) There exists $M > 0$ such that $|I(h) - 2f(0)h| \leq Mh^2$ for any $h \in [0, 1]$.

b) The sequence $(a_n)_{n \geq 1}$ defined by $a_n = \sum_{k=1}^n \sqrt{k} \left| I\left(\frac{1}{k}\right) \right|$ is convergent if and only if $f(0) = 0$.

148. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function, two times derivable on $\mathbb{R}$, for which there exists $c \in \mathbb{R}$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

for all $a \neq b \in \mathbb{R}$. Prove that $f''(c) = 0$.

149. a) Consider a differentiable and convex function $f : [0, \infty) \rightarrow [0, \infty)$. Show that if $f(x) \leq x$ for every $x \geq 0$, then $f'(x) \leq 1$ for every $x \geq 0$.

b) Determine differentiable and convex functions $f : [0, \infty) \rightarrow [0, \infty)$ which have the property that $f(0) = 0$ and $f'(x)f(f(x)) = x$ for every $x \geq 0$.

150. Let $f : \mathbb{R} \rightarrow [0, \infty)$ be a twice differentiable function whose second derivative is continuous, such that

$$\lim_{x \rightarrow \infty} [f(x) + f'(x) + f''(x)] = C.$$

Prove that $\lim_{x \rightarrow \infty} f(x) = C$.

151. Let f be a real function with a continuous third derivative such that $f(x)$, $f'(x)$, $f''(x)$, and $f'''(x)$ are positive for all x . Suppose that $f'''(x) \leq f(x)$ for all x . Show that $f'(x) < 2f(x)$ for all x .

152. Suppose that f is a function on the interval $[1, 3]$ such that $-1 \leq f(x) \leq 1$ for all x and $\int_1^3 f(x) dx = 0$. How large can $\int_1^3 \frac{f(x)}{x} dx$ be?

153. a) Prove that there exists a differentiable function $f : (0, \infty) \rightarrow (0, \infty)$ such that $f(f'(x)) = x$ for all $x > 0$.

b) Prove that there is no differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f'(x)) = x$ for all $x \in \mathbb{R}$.

154. Let $a > 0$ and let $\mathcal{F} = \{f : [0, 1] \rightarrow \mathbb{R} \mid f \text{ is concave and } f(0) = 1\}$. Determine

$$\min_{f \in \mathcal{F}} \left\{ \left(\int_0^1 f(x) dx \right)^2 - (a+1) \int_0^1 x^{2a} f(x) dx \right\}.$$

155. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that there exists $M > 0$ satisfying

$$|f(x_1 + \cdots + x_n) - f(x_1) - \cdots - f(x_n)| \leq M, \quad \forall x_1, \dots, x_n \in \mathbb{R}.$$

Prove that $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$.

156. Let $f : (0, 1) \rightarrow \mathbb{R}$ be a function satisfying $\lim_{x \rightarrow 0^+} f(x) = 0$. If there exists $\lambda \in (0, 1)$ such that $\lim_{x \rightarrow 0^+} \frac{|f(x) - f(\lambda x)|}{x} = 0$, prove that

$$\lim_{x \rightarrow 0^+} \frac{f(x)}{x} = 0.$$

157. Determine twice differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which verify the relation

$$(f'(x))^2 + f''(x) \leq 0, \quad \forall x \in \mathbb{R}.$$

158. For an arbitrary number $x_0 \in (0, \pi)$, define recursively the sequence $(x_n)_{n \geq 0}$ by $x_{n+1} = \sin(x_n)$. Compute

$$\lim_{n \rightarrow \infty} \sqrt{n} x_n.$$

159. Let a be a real number, and $f : \mathbb{R} \rightarrow \mathbb{R}$ a non-decreasing function. Prove that f is continuous at a if and only if there exists a sequence $(a_n)_{n \geq 1}$ of positive real numbers such that

$$\int_a^{a+a_n} f(x) dx + \int_a^{a-a_n} f(x) dx \leq \frac{a_n}{n}$$

for all natural numbers n .

160. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a surjective function that has the property that if the sequence $(f(x_n))_{n \geq 1}$ is convergent, then the sequence $(x_n)_{n \geq 1}$ is convergent. Prove that f is continuous.

161. Let $f : [0, \infty) \rightarrow (0, \infty)$ be a continuous function.

a) Show that there exists a natural number n_0 and a sequence of positive real numbers $(x_n)_{n \geq n_0}$ satisfying

$$n \int_0^{x_n} f(t) dt = 1, \quad \forall n > n_0.$$

b) Prove that the sequence $(nx_n)_{n \geq n_0}$ is convergent and find its limit.

162. Let $f : [a, b] \rightarrow \mathbb{R}$ be a function with the Intermediate Value Property such that $f(a) * f(b) < 0$. Show that there exist α, β such that $a < \alpha < \beta < b$ and

$$f(\alpha) + f(\beta) = f(\alpha) * f(\beta).$$

163. Let $f : [0, 1] \rightarrow (0, 1)$ be a surjective function.

a) Prove that f has at least one point of discontinuity.

b) Given that f admits a limit at any point of the interval $[0, 1]$, show that it has at least two points of discontinuity.

164. Let $f : [0, \infty) \rightarrow (0, \infty)$ be an increasing function and $g : [0, \infty) \rightarrow \mathbb{R}$ a twice differentiable function such that g'' is continuous and

$$g''(x) + f(x)g(x) = 0, \quad \forall x \geq 0.$$

a) Provide an example of such functions, with $g \not\equiv 0$.

b) Prove that g is bounded.

165. Find the number of trapeziums that can be formed using the vertices of a regular n -sided polygon.

166. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous function, constant on $\mathbb{Z}_{\geq 0}$. For any $0 \leq a < b < c < d$ which satisfy $f(a) = f(c)$ and $f(b) = f(d)$, we also have

$$f\left(\frac{a+b}{2}\right) = f\left(\frac{c+d}{2}\right).$$

Prove that f is a constant function.

167. Let f_n be a sequence of real-valued C^1 functions on $[0, 1]$ such that for all $n \in \mathbb{N}$ the following hold:

$$|f'_n(x)| \leq \frac{1}{\sqrt{x}} \quad (0 < x \leq 1), \quad \int_0^1 f_n(x) dx = 0.$$

Prove that f_n has a convergent subsequence that converges uniformly on $[0, 1]$.

168. Let $\chi_{\mathbb{Q}}$ denote the characteristic function of the rationals in $[0, 1]$. Does there exist a sequence of continuous functions $f_n : [0, 1] \rightarrow \mathbb{R}$ such that f_n converges to $\chi_{\mathbb{Q}}$ pointwise?

169. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that

$$\int_0^1 f(t) dt = \int_0^1 tf(t) dt.$$

Prove that there exists $c \in (0, 1)$ such that

$$\int_0^c f(t) dt = \frac{c}{2} f(c).$$

170. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^1 f(t) dt = \int_0^1 tf(t) dt$. Prove that there exists $c \in (0, 1)$ such that

$$cf(c) = 2 \int_c^1 f(t) dt.$$

171. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that

$$f'(x) = f^2(x)f(-x).$$

Find an explicit formula for f .

172. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^1 f(t) dt = 1$ and

$$\int_0^1 (1 - f(x))e^{-f(x)} dx \leq 0.$$

Prove that $f(x) = 1$ for all $x \in [0, 1]$.

173. Let $f : [a, b] \rightarrow [0, +\infty)$ be a continuous and not everywhere 0 function. Prove that

$$\lim_{n \rightarrow +\infty} \frac{\int_a^b f^{n+1}(t) dt}{\int_a^b f^n(t) dt} = \sup_{x \in [a, b]} f(x).$$

174. Examine if there exists a continuous function $f : [1, +\infty) \rightarrow \mathbb{R}$ such that $f(x) > 0$ for all $x \in [1, +\infty)$ and $\int_1^\infty f(t) dt$ converges whereas $\int_1^\infty f^2(t) dt$ diverges.

175. Let $0 < x_1 < 1$ and $x_{n+1} = x_n(1 - x_n)$ for $n = 1, 2, 3, \dots$. Show that

$$\lim_{n \rightarrow \infty} nx_n = 1.$$

176. a) Let $(x_n)_{n \geq 1}$ be a sequence such that $x_1 \in (0, 1)$ and $x_{n+1} = x_n(1 - x_n^2)$ for $n \geq 1$. Evaluate

$$\lim_{n \rightarrow \infty} \sqrt{n} x_n.$$

b) Let $(x_n)_{n \geq 1}$ be a sequence such that $x_1 = x > 1$ and $x_{n+1} = x_n + \sqrt{x_n - 1}$ for $n \geq 1$. Evaluate

$$\lim_{n \rightarrow \infty} \frac{4x_n - n^2}{n \log n}.$$

177. Consider the harmonic sequence $(H_n)_{n \geq 1}$, $H_n = 1 + \frac{1}{2} + \dots + \frac{1}{n}$, $n \geq 1$. Show that the sequence $x_n = \{H_n\}$ diverges, where $\{x\}$ denotes the fractional part of the real number x .

178. Let $(a_n)_{n \geq 1}$ be a decreasing sequence of positive reals such that $\sum_{n=1}^\infty a_n$ converges. Show that $\lim_{n \rightarrow \infty} na_n = 0$.

179. Evaluate the following limits:

a) $\lim_{n \rightarrow \infty} \left(\frac{\left(1 + \frac{1}{n}\right)^n}{e} \right)^n$;

b) $\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(\frac{k}{k+n} \right)^{e^{1999/n} - 1}$.

180. Define the sequence $x_1, x_2, \dots$ inductively by $x_1 = \sqrt{5}$ and $x_{n+1} = x_n^2 - 2$ for each $n \geq 1$. Compute

$$\lim_{n \rightarrow \infty} \frac{x_1 x_2 \cdots x_n}{x_{n+1}}.$$

181. Suppose that $a_0 = 1$ and that $a_{n+1} = a_n + e^{-a_n}$ for $n = 0, 1, 2, \dots$. Does the sequence $(a_n - \log n)_{n \geq 1}$ have a finite limit?

182. Let k be an integer greater than 1. Suppose that $a_0 > 0$, and define the sequence $(a_n)_{n \geq 0}$ by

$$a_{n+1} = a_n + \frac{1}{\sqrt[k]{a_n}}, \quad n \geq 0.$$

Evaluate

$$\lim_{n \rightarrow \infty} \frac{a_n^{k+1}}{n^k}.$$

183. a) Does there exist a sequence $(a_n)_{n \geq 1}$ of positive reals such that the series $\sum_{n=1}^{\infty} \frac{1}{a_n}$

converges, and $\prod_{k=1}^n a_k \leq n^n$ for all $n \geq 1$?

b) Does there exist a sequence $(a_n)_{n \geq 1}$ of positive reals such that $\sum_{k=1}^n \frac{1}{a_k} \leq 2008$, and $\sum_{k=1}^n a_k \leq n^2$ for all $n \geq 1$?

184. Let $(a_n)_{n \geq 1}$ be a sequence of positive reals such that the series $\sum_{n=1}^{\infty} a_n$ converges. Show

that the series $\sum_{n=1}^{\infty} a_n^{n/(n+1)}$ converges also.

185. Let $(a_n)_{n \geq 1}$ be a decreasing sequence of positive reals. Let $s_n = a_1 + a_2 + \cdots + a_n$ and

$$b_n = \frac{1}{a_{n+1}} - \frac{1}{a_n}, \quad n \geq 1.$$

Prove that if $(s_n)_{n \geq 1}$ is convergent, then $(b_n)_{n \geq 1}$ is unbounded.

186. Let $(x_n)_{n \geq 2}$ be a sequence of real numbers such that $x_2 > 0$ and

$$x_{n+1} = -1 + \sqrt[n]{1 + nx_n}, \quad n \geq 2.$$

Show that:

a) $\lim_{n \rightarrow \infty} x_n = 0$;

b) $\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = 1$ and $\lim_{n \rightarrow \infty} nx_n = 0$.

187. Let $(a_n)_{n \geq 1}$ be a sequence of positive reals. Show that

$$\limsup_{n \rightarrow \infty} n \left(\frac{1 + a_{n+1}}{a_n} - 1 \right) \geq 1,$$

and

$$\limsup_{n \rightarrow \infty} \left(\frac{a_1 + a_{n+1}}{a_n} \right)^n \geq e.$$

188. Let $(a_n)_{n \geq 1}$ be a sequence such that $x_n = \sum_{k=1}^n a_k^2$ converges, while $y_n = \sum_{k=1}^n a_k$ is unbounded. Show that the sequence $(b_n)_{n \geq 1}$, $b_n = \{y_n\}$, diverges, where $\{x\}$ denotes the fractional part of the real number x .

189. Show that if the series $\sum_{n=1}^{\infty} \frac{1}{p_n}$ is convergent, where $p_1, p_2, \dots, p_n$ are positive real numbers, then the series

$$\sum_{n=1}^{\infty} \frac{n^2}{(p_1 + p_2 + \cdots + p_n)^2} \cdot p_n$$

is also convergent.

190. Determine, with proof, whether the series $\sum_{n=1}^{\infty} \frac{1}{n^{1.8 + \sin n}}$ converges or diverges.

191. (Carleman's inequality) Let $a_1, a_2, \dots, a_n$ be a sequence of nonnegative real numbers. Prove that

$$\sum_{n=1}^{\infty} \sqrt[n]{a_1 a_2 \cdots a_n} \leq e \sum_{n=1}^{\infty} a_n.$$

192. (Hardy's inequality) Let $a_1, a_2, \dots, a_n$ be a sequence of nonnegative real numbers and $p > 1$. Prove that

$$\sum_{n=1}^{\infty} \left(\frac{a_1 + a_2 + \cdots + a_n}{n} \right)^p \leq \left(\frac{p}{p-1} \right)^p \sum_{n=1}^{\infty} a_n^p.$$

193. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a function satisfying $f(x)e^{f(x)} = x$ for $x \geq 0$. Prove that:

- a) f is monotone;
 - b) $\lim_{x \rightarrow \infty} f(x) = \infty$;
 - c) $\frac{f(x)}{\log x}$ tends to 1 as $x \rightarrow \infty$.
-

194. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a function satisfying

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x^k} = a, \quad k \neq 1,$$

and

$$\lim_{x \rightarrow \infty} \frac{f(x+1) - f(x)}{x^{k-1}} = b \in \mathbb{R}.$$

Prove that $b = ka$.

195. Let $f : (a, b) \rightarrow \mathbb{R}$ be a function such that $\lim_{x \rightarrow x_0} f(x)$ exists for any $x_0 \in [a, b]$. Prove that f is bounded if and only if for all $x_0 \in [a, b]$, $\lim_{x \rightarrow x_0} f(x)$ is finite.

196. Let $a, b \in \left(0, \frac{1}{2}\right)$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f(f(x)) = af(x) + bx,$$

for all x . Prove that $f(0) = 0$ and find all such functions.

197. Prove that there is no continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) \in \mathbb{Q} \implies f(x+1) \in \mathbb{R} \setminus \mathbb{Q}$.

198. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f(x) \leq f\left(x + \frac{1}{n}\right),$$

for every real x and positive integer n . Show that f is nondecreasing.

199. Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ subject to the conditions:

- (i) $f(f(f(x))) + 2x = f(3x)$ for all $x > 0$;
 - (ii) $\lim_{x \rightarrow \infty} (f(x) - x) = 0$.
-

200. Let $f(x)$ be a continuous function such that $f(2x^2 - 1) = 2xf(x)$ for all x . Show that $f(x) = 0$ for $-1 \leq x \leq 1$.

201. Calculate

$$\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right) \ln\left(1 + \frac{1}{2n}\right) \ln\left(1 + \frac{1}{2n+1}\right).$$

202. Let $f(x) = \sum_{k=1}^n a_k \sin kx$, with $a_1, a_2, \dots, a_n$ real numbers and $n \geq 1$. Prove that if $f(x) \leq |\sin x|$ for all $x \in \mathbb{R}$, then

$$\left| \sum_{k=1}^n k a_k \right| \leq 1.$$

203. Let $f : [a, b] \rightarrow \mathbb{R}$ be a function, continuous on $[a, b]$ and twice differentiable on (a, b) . If $f(a) = f(b)$ and $f'(a) = f'(b)$, prove that for every real number λ , the equation

$$f''(x) - \lambda(f'(x))^2 = 0$$

has at least one zero in the interval (a, b) .

204. Let $\alpha > 1$ be a real number, and let $(u_n)_{n \geq 1}$ be a sequence of positive numbers such that $\lim_{n \rightarrow \infty} u_n = 0$ and $\lim_{n \rightarrow \infty} \frac{u_n - u_{n+1}}{u_n^\alpha}$ exists and is nonzero. Prove that $\sum_{n=1}^{\infty} u_n$ converges if and only if $\alpha < 2$.

205. Prove that, for any two bounded functions $g_1, g_2 : \mathbb{R} \rightarrow [1, \infty)$, there exist functions $h_1, h_2 : \mathbb{R} \rightarrow \mathbb{R}$ such that for every $x \in \mathbb{R}$,

$$\sup_{s \in \mathbb{R}} (g_1(s)^x g_2(s)) = \max_{t \in \mathbb{R}} (x h_1(t) + h_2(t)).$$

206. Is there a strictly increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f'(x) = f(f(x))$ for all reals x ?

207. Does there exist a continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(x) > 0$ and $f'(x) = f(f(x))$ for all reals x ?

208. Find all differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f'(x) = \frac{f(x+n) - f(x)}{n}$$

for all real numbers x and all positive integers n .

209. Find the set of all real numbers k with the following property: for any positive, differentiable function f that satisfies $f'(x) > f(x)$ for all x , there is some number N such that $f(x) > e^{kx}$ for all $x > N$.

210. Let $f : (1, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that

$$f'(x) = \frac{x^2 - (f(x))^2}{x^2 ((f(x))^2 + 1)} \quad \text{for all } x > 1.$$

Prove that $\lim_{x \rightarrow \infty} f(x) = \infty$.

211. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} x & \text{if } x \leq e \\ x f(\ln x) & \text{if } x > e \end{cases}$$

Does $\sum_{n=1}^{\infty} \frac{1}{f(n)}$ converge?

212. Find all sequences $a_0, a_1, \dots, a_n$ of real numbers with $a_n \neq 0$, for which the following statement is true: if $f : \mathbb{R} \rightarrow \mathbb{R}$ is an n times differentiable function and $x_0 < x_1 < \dots < x_n$ are real numbers such that $f(x_0) = f(x_1) = \dots = f(x_n) = 0$, then there is $h \in (x_0, x_n)$ for which

$$a_0 f(h) + a_1 f'(h) + \dots + a_n f^{(n)}(h) = 0.$$

213. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function. Assume that

$$\lim_{x \rightarrow \infty} \left(f(x) + \frac{f'(x)}{x} \right) = 0.$$

Prove that $\lim_{x \rightarrow \infty} f(x) = 0$.

214. Let f be a twice continuously differentiable function on $(0, \infty)$ such that $\lim_{x \rightarrow 0^+} f'(x) = -\infty$ and $\lim_{x \rightarrow 0^+} f''(x) = +\infty$. Show that

$$\lim_{x \rightarrow 0^+} \frac{f(x)}{f'(x)} = 0.$$

215. Determine all Riemann integrable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\int_0^{x+1/n} f(t) dt = \int_0^x f(t) dt + \frac{1}{n} f(x),$$

for all reals x and all positive integers n .

216. For each continuous function $f : [0, 1] \rightarrow \mathbb{R}$, let $I(f) = \int_0^1 x^2 f(x) dx$ and $J(f) = \int_0^1 x f^2(x) dx$. Find the maximum value of $I(f) - J(f)$ over all such functions.

217. Let $f : [0, 1] \rightarrow [0, \infty)$ be a continuous function. Show that

$$\int_0^1 f^3(x) dx \geq 4 \int_0^1 x^2 f(x) dx \int_0^1 x f^2(x) dx.$$

218. Find all differentiable functions $f : [0, 1] \rightarrow \mathbb{R}$ with continuous derivative such that $f(0) = -\frac{1}{6}$ and

$$\int_0^1 (f'(x))^2 dx \leq 2 \int_0^1 f(x) dx.$$

219. a) Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuously differentiable and satisfy $f(1) = 0$. Show that

$$\int_0^1 |f(x)|^2 dx \leq 4 \int_0^1 x^2 (f'(x))^2 dx.$$

b) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative such that $f(1) = 0$. Show that

$$4 \int_0^1 x^2 |f'(x)|^2 dx \geq \int_0^1 |f(x)|^2 dx + \left(\int_0^1 f(x) dx \right)^2.$$

220. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function such that

$$\int_0^1 f(x^n) dx \leq \int_0^1 f^n(x) dx, \quad n \geq 1.$$

Show that $f(x) = 0$ for all $x \in [0, 1]$.

221. Find all continuous functions $f : \mathbb{R} \rightarrow [1, \infty)$ for which there exists $a \in \mathbb{R}$ and a positive integer k such that

$$f(x)f(2x) \cdots f(nx) \leq an^k,$$

for every real number x and positive integer n .

222. Let $f : [0, 1] \rightarrow \mathbb{R}^+$ be an integrable function. Compute the following limits:

- a) $\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{1}{n} f\left(\frac{k}{n}\right) \right);$
b) $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \exp\left(\frac{1}{n} f\left(\frac{k}{n}\right)\right) - n \right).$

223. Prove that

$$\lim_{n \rightarrow \infty} n \left(\frac{\pi}{4} - n \int_0^1 \frac{x^n}{1+x^{2n}} dx \right) = \int_0^1 f(x) dx,$$

where $f(x) = \frac{\arctan x}{x}$ if $x \in (0, 1]$ and $f(0) = 1$.

224. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^1 f(x) dx = 0$. Prove that there is $c \in (0, 1)$ such that

$$\int_0^c x f(x) dx = 0.$$

225. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a continuous function. Show that:

- a) if $\int_0^1 f(\sin(x + \alpha)) dx = 0$ for every $\alpha \in \mathbb{R}$, then $f(x) = 0$ for all $x \in [-1, 1]$;
b) if $\int_0^1 f(\sin(nx)) dx = 0$ for every $n \in \mathbb{Z}$, then $f(x) = 0$ for all $x \in [-1, 1]$.
-

226. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a periodic function, with period 1, integrable on $[0, 1]$. For a strictly increasing and unbounded sequence $(x_n)_{n \geq 0}$, $x_0 = 0$, with $\lim_{n \rightarrow \infty} (x_{n+1} - x_n) = 0$, denote $r(n) = \max\{k \mid x_k \leq n\}$.

a) Show that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^{r(n)} (x_k - x_{k+1}) f(x_k) = \int_0^1 f(x) dx.$$

b) Show that

$$\lim_{n \rightarrow \infty} \frac{1}{\log n} \sum_{k=1}^{r(n)} \frac{f(\log k)}{k} = \int_0^1 f(x) dx.$$

227. Let f and g be two continuous, distinct functions from $[0, 1]$ to $(0, +\infty)$ such that $\int_0^1 f(x) dx = \int_0^1 g(x) dx$. Let $(y_n)_{n \geq 1}$ be the sequence defined by

$$y_n = \int_0^1 \frac{f^{n+1}(x)}{g^n(x)} dx.$$

Prove that (y_n) is an increasing and divergent sequence.

228. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous and bounded function such that

$$x \int_x^{x+1} f(t) dt = \int_0^x f(t) dt, \quad \text{for any } x \in \mathbb{R}.$$

Prove that f is a constant function.

229. Let $f : [0, 1] \rightarrow \mathbb{R}$ be an integrable function such that $0 < \left| \int_0^1 f(x) dx \right| \leq 1$. Show that there exist $x_1 \neq x_2$ in $[0, 1]$ such that

$$\int_{x_1}^{x_2} f(x) dx = (x_1 - x_2)^{2002}.$$

230. Let $f : \mathbb{R} \rightarrow (0, \infty)$ be a continuously differentiable function. Prove that

$$\left| \int_0^1 f^3(x) dx - f^2(0) \int_0^1 f(x) dx \right| \leq \max_{[0,1]} |f'(x)| \left(\int_0^1 f(x) dx \right)^2.$$

231. Let $f : [0, 1] \rightarrow [0, \infty)$ be an integrable function. Show that

$$\int_0^1 f(x) dx \int_0^1 f^2(x) dx \leq \int_0^1 f^3(x) dx,$$

and

$$\int_0^1 x f(x) dx \int_0^1 x^2 f(x) dx \leq \int_0^1 f(x) dx \int_0^1 x^3 f(x) dx.$$

232. Show that for any continuous function $f : [0, 1] \rightarrow \mathbb{R}$,

$$\frac{1}{4} \int_0^1 f^2(x) dx + 2 \left(\int_0^1 f(x) dx \right)^2 \geq 3 \int_0^1 f(x) dx \int_0^1 x f(x) dx.$$

233. Let $f : [0, 1] \rightarrow \mathbb{R}$ be an integrable function such that

$$\int_0^1 f(x) dx = \int_0^1 xf(x) dx = \int_0^1 x^2 f(x) dx = 1.$$

Show that $\int_0^1 f^2(x) dx \geq 9$.

234. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function with continuous derivative such that

$$\int_0^1 f(x) dx = \int_0^1 xf(x) dx = 1.$$

Show that $\int_0^1 (f'(x))^2 dx \geq 30$.

235. a) Consider $f : [0, \infty) \rightarrow \mathbb{R}$ and $g : [0, 1] \rightarrow \mathbb{R}$ two continuous functions such that $\lim_{x \rightarrow \infty} f(x) = L \in \mathbb{R}$. Show that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \int_0^n f(x) g\left(\frac{x}{n}\right) dx = L \int_0^1 g(x) dx.$$

b) Find $\lim_{n \rightarrow \infty} \frac{1}{n} \int_0^n \frac{x \log(1 + x/n)}{1 + x} dx$.

236. a) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^1 f(x) dx = 0$. Show that there exists $c \in (0, 1)$ such that

$$f(c) = \int_0^c f(x) dx.$$

b) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $f(1) = 0$. Show that there exists $c \in (0, 1)$ such that

$$f(c) = \int_0^c f(x) dx.$$

237. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function for which there exist $a, b \in (0, 1)$, $a < b$, such that $\int_0^a f(x) dx = 0$ and $\int_b^1 f(x) dx = 0$. Let $M = \sup_{x \in [0, 1]} |f'(x)|$. Show that

$$\left| \int_0^1 f(x) dx \right| \leq \frac{1 - a + b}{4} M.$$

Moreover, if $f : [0, 1] \rightarrow [0, \infty)$ is differentiable on $[0, 1]$ and there exists $a \in (0, 1]$ with $\int_0^a f(x) dx = 0$, show that

$$\left| \int_0^1 f(x) dx \right| \leq \frac{1-a}{2} \sup_{x \in (0,1)} |f'(x)|.$$

238. a) Let $f_0 : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Define the sequence of functions $f_n : [0, 1] \rightarrow \mathbb{R}$ by $f_n(x) = \int_0^x f_{n-1}(t) dt$ for all integers $n \geq 1$. Prove that the series $\sum_{n=1}^{\infty} f_n(x)$ is convergent for every $x \in [0, 1]$ and find an explicit formula for the sum of the series.

b) Let $F_0 = \ln x$. For $n \geq 0$ and $x > 0$, let $F_{n+1}(x) = \int_0^x F_n(t) dt$. Evaluate $\lim_{n \rightarrow \infty} \frac{n! F_n(1)}{\ln n}$.

239. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a strictly decreasing continuous function such that $\lim_{x \rightarrow \infty} f(x) = 0$. Prove that $\int_0^{\infty} \frac{f(x) - f(x+1)}{f(x)} dx$ diverges.

240. Let $(a_n)_{n \geq 1}$ be defined by

$$a_n = \frac{3}{2} \log n - \int_0^1 \log(1^t + 2^t + \cdots + n^t) dt.$$

i) Show that the sequence a_n is convergent and find its limit a .

ii) Show that $0 < n(a - a_n) < \frac{3}{2}$.

241. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous and periodic function of period $T > 0$. Show that:

i) $\lim_{x \rightarrow \infty} \frac{1}{x} \int_0^x f(t) dt = \frac{1}{T} \int_0^T f(t) dt.$

ii) $\lim_{n \rightarrow \infty} \int_a^b f(nx) dx = \frac{b-a}{T} \int_a^b f(t) dt.$

iii) Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions such that $f(x+1) = f(x)$ and $g(x+1) = g(x)$ for all real x . Prove that

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) g(nx) dx = \int_0^1 f(x) dx \int_0^1 g(x) dx.$$

242. Let $a_1, a_2, \dots$ be real numbers. Suppose there is a constant A such that for all n ,

$$\int_{-\infty}^{\infty} \left(\sum_{i=1}^n \frac{1}{1 + (x - a_i)^2} \right)^2 dx \leq An.$$

Prove there is a constant $B > 0$ such that for all n ,

$$\sum_{i,j=1}^n (1 + (a_i - a_j)^2) \geq Bn^3.$$

243. a) (Hermite–Hadamard) Let $f : [a, b] \rightarrow \mathbb{R}$ be a convex function. Show that

$$(b - a)f\left(\frac{a + b}{2}\right) \leq \int_a^b f(x) dx \leq (b - a)\frac{f(a) + f(b)}{2}.$$

b) Show that $\int_k^{k+1} \log x dx \geq \frac{\log k + \log(k+1)}{2}$ for $k \geq 1$, and $\int_{k-1/2}^{k+1/2} \log x dx \leq \log k$ for $k \geq 1$.

c) Consider the sequence $(a_n)_{n \geq 1}$ defined by

$$a_n = \int_1^n \log x dx - \log 2 - \dots - \log(n-1) - \frac{1}{2} \log n, \quad n \geq 1.$$

Show that a_n is increasing and $0 \leq a_n \leq \frac{1}{2} \log \frac{5}{4}$.

d) Prove that $\sqrt[4]{\frac{4}{5}} e \sqrt{n} \left(\frac{n}{e}\right)^n \leq n! \leq e \sqrt{n} \left(\frac{n}{e}\right)^n$ for $n \geq 1$.

e) (Stirling) Show that $\lim_{n \rightarrow \infty} \frac{n!}{\left(\frac{n}{e}\right)^n \sqrt{2\pi n}} = 1$.

244. Show that for any continuous function $f : [-1, 1] \rightarrow \mathbb{R}$ we have the inequalities

$$\int_{-1}^1 f^2(x) dx \geq \frac{1}{2} \left(\int_{-1}^1 f(x) dx \right)^2 + \frac{3}{2} \left(\int_{-1}^1 x f(x) dx \right)^2,$$

and

$$\int_{-1}^1 f^2(x) dx \geq \frac{5}{2} \left(\int_{-1}^1 x^2 f(x) dx \right)^2 + \frac{3}{2} \left(\int_{-1}^1 x f(x) dx \right)^2.$$

245. a) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Show that $\lim_{n \rightarrow \infty} \int_0^1 x^n f(x) dx = 0$.

b) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Show that

$$\lim_{n \rightarrow \infty} n \int_0^1 x^n f(x) dx = f(1), \quad \lim_{n \rightarrow \infty} n \int_0^1 x^n f(x^n) dx = \int_0^1 f(x) dx.$$

c) Let $f, g : [0, 1] \rightarrow \mathbb{R}$ be two continuous functions. Show that

$$\lim_{n \rightarrow \infty} \frac{\int_0^1 x^n f(x) dx}{\int_0^1 x^n g(x) dx} = \frac{f(1)}{g(1)}, \quad \lim_{n \rightarrow \infty} \frac{\int_0^1 x^n f(x^n) dx}{\int_0^1 x^n g(x^n) dx} = \frac{\int_0^1 f(x) dx}{\int_0^1 g(x) dx}.$$

d) Find a real number c and a positive number L for which

$$\lim_{r \rightarrow \infty} r^c \frac{\int_0^{\pi/2} x^r \sin x dx}{\int_0^{\pi/2} x^r \cos x dx} = L.$$

246. Let $(a_n)_{n \in \mathbb{N}}$ be the sequence defined by

$$a_0 = 1, \quad a_{n+1} = \frac{1}{n+1} \sum_{k=0}^n \frac{a_k}{n-k+2}.$$

Find the limit $\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{a_k}{2^k}$.

247. Prove that for any real numbers $a_1, a_2, \dots, a_n$,

$$\sum_{1 \leq i, j \leq n} \frac{ij}{i+j-1} a_i a_j \geq \left(\sum_{i=1}^n a_i \right)^2.$$

248. (Hardy) Let $p > 1$ and let $f : [0, \infty) \rightarrow \mathbb{R}$ be a differentiable increasing function such that $f(0) = 0$ and $\int_0^\infty (f'(x))^p dx$ is finite. Show that

$$\int_0^\infty x^{-p} |f(x)|^p dx \leq \left(\frac{p}{p-1} \right)^p \int_0^\infty (f'(x))^p dx.$$

249. Let $u_1, u_2, \dots, u_n \in C([0, 1]^n)$ be nonnegative and continuous functions, where u_j does not depend on the j -th variable for $j = 1, 2, \dots, n$. Show that

$$\left(\int_{[0,1]^n} \prod_{j=1}^n u_j \right)^{n-1} \leq \prod_{j=1}^n \int_{[0,1]^n} u_j^{n-1}.$$

250. (Zagier) Prove that if $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ are nonnegative, then

$$\left(\sum_{1 \leq i, j \leq n} \min(a_i, a_j) \right) \left(\sum_{1 \leq i, j \leq n} \min(b_i, b_j) \right) \geq \left(\sum_{1 \leq i, j \leq n} \min(a_i, b_j) \right)^2.$$

251. Let $\{D_1, D_2, \dots, D_n\}$ be a set of disks in the Euclidean plane and $a_{ij} = S(D_i \cap D_j)$ be the area of $D_i \cap D_j$. Prove that for any real numbers $x_1, x_2, \dots, x_n$,

$$\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j \geq 0.$$

252. Let $a, b \in \mathbb{R}$, $f : [a, b] \rightarrow [0, 1]$ an increasing function, and define the sequence $(c_n)_{n \geq 1}$

by $c_n = n \int_a^b f^n(x) dx$. Show that:

i) $\lim_{n \rightarrow \infty} (c_{n+1} - c_n) = 0$.

ii) $\lim_{n \rightarrow \infty} \frac{c_{n+1}}{c_n} = f(b^-)$.

253. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a C^1 function. Show that

$$\lim_{n \rightarrow \infty} n \left[\frac{1}{2} \left(\int_0^1 f(x) dx \right)^2 - \frac{1}{n^2} \sum_{1 \leq i, j \leq n} f\left(\frac{i}{n}\right) f\left(\frac{j}{n}\right) \right] = \frac{1}{2} \int_0^1 f^2(x) dx + \frac{f(0) - f(1)}{2} \int_0^1 f(x) dx.$$

254. a) Find the limit $\lim_{n \rightarrow \infty} \left(\int_0^1 (1 + x^n)^n dx \right)^{1/n}$.

b) Prove that $\lim_{n \rightarrow \infty} n^2 \left(\int_0^1 \sqrt[n]{1 + x^n} dx - 1 \right) = \frac{\pi^2}{12}$.

255. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function such that f' is bounded. Show that

$$\int_0^1 f^2(x) dx - \left(\int_0^1 f(x) dx \right)^2 \leq \frac{1}{12} \left(\sup_{x \in [0,1]} |f'(x)| \right)^2.$$

256. Assume that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous injection. Prove that if there exists n such that the n -th iterate of f is the identity, that is, $f^n(x) = x$ for all $x \in \mathbb{R}$, then:

- a) $f(x) = x$ for all $x \in \mathbb{R}$, if f is strictly increasing;
 - b) $f^2(x) = x$ for all $x \in \mathbb{R}$, if f is strictly decreasing.
-

257. (Continuity of square-root-type functional equations)

- a) Assume that $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x)^2 = -x$ for all $x \in \mathbb{R}$. Show that f cannot be continuous.
 - b) Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f^2(x) = \cos(x)$ for all $x \in \mathbb{R}$.
-

258. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which have the Intermediate Value Property and such that there is $n \in \mathbb{N}$ for which $f^n(x) = -x$ for all $x \in \mathbb{R}$.

259. Give an example of a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ which attains each of its values exactly three times. Does there exist a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ which attains each of its values exactly two times?

260. Let $f \in C^2[0, 1]$ be such that $f(0) = f(1)$. Show that

$$3f'(1)^2 \leq \int_0^1 f''(x)^2 dx.$$

261. (Monotonic functions and continuity)

- a) If f is monotonic on $[a, b]$, then the set of discontinuities of f is countable.
 - b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a surjective monotonous function. Show that f must be continuous.
-

262. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function. Consider any sequence given by $x_{n+1} = f(x_n)$. Show that (x_n) converges if and only if $\lim_{n \rightarrow \infty} (x_{n+1} - x_n) = 0$.

263. Let f be a three times continuously differentiable function with the property that $f(0) = f'(0) = 0 < f''(0)$. Define

$$g(x) = \frac{d}{dx} \left(\frac{\sqrt{f(x)}}{f'(x)} \right)$$

for $x \neq 0$ and $g(0) = 0$. Show that g is bounded in a neighbourhood of 0. Does this conclusion still follow if f is only twice continuously differentiable?

264. Let $f : [0, 1] \rightarrow \mathbb{R}$ be given by $f(x) = 2x(1 - x)$. Denote $f_n = f \circ \cdots \circ f$ for the n -fold composition of f . Determine

$$\int_0^1 f_n(x) dx.$$

Does this sequence converge?

265. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function satisfying $f(0) = 2$, $f'(0) = -2$ and $f(1) = 1$. Prove that there exists some $c \in (0, 1)$ such that

$$f(c) \cdot f'(c) + f''(c) = 0.$$

266. Let $c \in (0, 1)$ and define

$$f(x) = \begin{cases} x/c & \text{if } x \in [0, c]; \\ (1 - x)/(1 - c) & \text{if } x \in [c, 1]. \end{cases}$$

Show that the equation $f^n(x) = x$ only has finitely many solutions for every $n \geq 1$, where f^n denotes the n -fold composition of f with itself. Show also that for every n , there are solutions to $f^n(x) = x$ for which this n is minimal.

267. (*) Let $f : (0, 1) \rightarrow [0, \infty)$ be a function that is non-zero only on the distinct points $a_1, a_2, \dots$. Suppose that

$$\sum_{n=1}^{\infty} f(a_n) < \infty.$$

Prove that f is differentiable at at least one $x \in (0, 1)$.

268. Let $f : [0, 1] \rightarrow [0, 1]$ be a strictly increasing function. Show that for some $x \in [0, 1]$, it holds that $f(x) = x$. What if f is strictly decreasing?

269. Let $f : [a, b] \rightarrow [a, b]$ be a continuous function and let $p \in [a, b]$. Define the sequence (p_n) by $p_0 = p$ and $p_{n+1} = f(p_n)$ for all $n \geq 0$. Suppose that the set $T_p = \{p_n \mid n \in \mathbb{N}\}$ is closed. Prove that T_p is finite.

270. (*) Let $g : [0, 1] \rightarrow \mathbb{R}$ be a function and define the sequence of functions $f_n : (0, 1] \rightarrow \mathbb{R}$ by setting $f_0(x) = g(x)$ and

$$f_{n+1}(x) = \frac{1}{x} \int_0^x f_n(t) dt$$

for all $x \in (0, 1]$ and $n \geq 0$. Determine $\lim_{n \rightarrow \infty} f_n(x)$ for every $x \in (0, 1]$.

271. Let $f, g : [a, b] \rightarrow [0, \infty)$ be continuous and non-decreasing functions such that for each $x \in [a, b]$, we have

$$\int_a^x \sqrt{f(t)} dt \leq \int_a^x \sqrt{g(t)} dt,$$

with equality for $x = b$. Prove that

$$\int_a^b \sqrt{1 + f(t)} dt \geq \int_a^b \sqrt{1 + g(t)} dt.$$

272. Let $f(x) = x^2 + bx + c$ where $b, c \in \mathbb{R}$. Now define the possibly empty set $M = \{x \in \mathbb{R} : |f(x)| < 1\}$. Clearly M is the union of disjoint open intervals. Let $|M|$ denote the sum of their lengths. Prove that $|M| \leq 2\sqrt{2}$.

273. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a three times differentiable function. Prove that for some $c \in (-1, 1)$, it holds that

$$f'''(c) = 3f(1) - 3f(-1) - 6f'(0).$$

274. Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for any real numbers $a < b$, the image $f([a, b])$ is a closed interval of length $b - a$.

275. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function with the property that for every $c > 0$, the graph of f can be transformed into the graph of $c \cdot f$ by translating and rotating. Does this imply that $f(x) = ax + b$ for some $a, b \in \mathbb{R}$?

276. Let $C \subset \mathbb{R}$ be a compact set and let $f : C \rightarrow C$ be a non-decreasing continuous function. Show that for some $p \in C$ it holds that $f(p) = p$.

277. Determine all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ with the property that

$$x - y \in \mathbb{Q} \implies f(x) - f(y) \in \mathbb{Q}.$$

278. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two functions with the property that $f(r) \leq g(r)$ for all $r \in \mathbb{Q}$. If f and g are non-decreasing, does this imply that $f(x) \leq g(x)$ for all $x \in \mathbb{R}$? What if f and g are both continuous?

279. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. We say some $x \in \mathbb{R}$ is a shadow-point of f if for some $y > x$ it holds that $f(y) > f(x)$. Let $a < b$ now be two real numbers that are not shadow-points of f , with the property that all points in (a, b) are shadow-points of f . Show that $f(x) \leq f(b)$ for all $x \in (a, b)$ and that $f(a) = f(b)$.

280. Compute

$$\lim_{A \rightarrow \infty} \frac{1}{A} \int_1^A A^{1/x} dx.$$

281. Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function that is differentiable on (a, b) . Suppose that f has infinitely many zeroes, but that there is no $x \in (a, b)$ such that $f(x) = f'(x) = 0$. Prove that $f(a)f(b) = 0$ and give an example of such a function on $[0, 1]$.

282. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous function satisfying the property that $\lim_{x \rightarrow \infty} f(x) = L$ exists. Prove that

$$\lim_{n \rightarrow \infty} \int_0^1 f(nx) dx = L.$$

283. (*) Define the sequence $f_1, f_2, \dots : [0, 1] \rightarrow \mathbb{R}$ of continuously differentiable functions by

$$f_1 = 1, \quad f'_{n+1} = f_n f_{n+1} \text{ on } (0, 1) \quad \text{and} \quad f_{n+1}(0) = 1.$$

Show that $\lim_{n \rightarrow \infty} f_n(x)$ exists for every $x \in [0, 1]$ and determine the limit function.

284. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions such that g is differentiable. Suppose that

$$(f(0) - g'(0))(g'(1) - f(1)) > 0.$$

Show that there exists some $c \in (0, 1)$ such that $f(c) = g'(c)$.

285. Let V be the set of all continuous functions $f : [0, 1] \rightarrow \mathbb{R}$, differentiable on $(0, 1)$, with the property that $f(0) = 0$ and $f(1) = 1$. Determine all $\alpha \in \mathbb{R}$ such that for every $f \in V$, there exists some $\xi \in (0, 1)$ such that $f(\xi) + \alpha = f'(\xi)$.

286. Determine all continuously differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying for all $x \in \mathbb{R}$ the equation

$$f(x)^2 = \int_0^x f(t)^2 + f'(t)^2 dt + 2023.$$

287. Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying for all $x, y \in \mathbb{R}$ the equation

$$f(x^2 - y^2) = (x - y)(f(x) + f(y)).$$

288. Consider a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(2x - f(x)) = x \quad \text{for all } x \in \mathbb{R}.$$

Suppose that $f(\alpha) = \alpha$ for some $\alpha \in \mathbb{R}$. Show that then $f(x) = x$ for all $x \in \mathbb{R}$. Does this conclusion still hold if f such an α need not necessarily exist?

289. Let $c > 0$. Determine all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ with the property that

$$f(x) = f(x^2 + c) \quad \text{for all } x \in \mathbb{R}.$$

290. Determine all $\alpha > 0$ with the property that for all differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}_{>0}$ such that $f'(x) > f(x)$ for all $x \in \mathbb{R}$, there exists some $N \in \mathbb{R}$ such that $f(x) > e^{\alpha x}$ for all $x > N$.

291. Let $a, b \in (0, 1/2)$ and let g be a continuous function satisfying the property that

$$g(g(x)) = ag(x) + bx \quad \text{for all } x \in \mathbb{R}.$$

Prove that $g(x) = cx$ for some $c \in \mathbb{R}$.

292. Prove that there exists no function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(f(x) + y) = f(x) + 3x + yf(y) \quad \text{for all } x, y \in \mathbb{R}.$$

293. Determine all differentiable functions $f : (0, \infty) \rightarrow (0, \infty)$ for which there exists some $a \in \mathbb{R}$ such that

$$f'(a/x) = x/f(x) \quad \text{for all } x > 0.$$

294. Determine all functions $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$ satisfying

$$xf(x^2f(y)) = yf(x) \quad \text{for all } x, y \in \mathbb{R}.$$

295. Determine all twice differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f'(x)f''(x) = 0$ for all $x \in \mathbb{R}$.

296. Determine all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(x) + \int_0^x (x-t)f(t) dt = 1 \quad \text{for all } x \in \mathbb{R}.$$

297. Let $f : [-2, 2] \rightarrow \mathbb{R}$ be continuously differentiable. Prove that there exists some $x \in (-2, 2)$ such that

$$f'(x) - f(x)^2 < 1.$$

298. (*) Let $f : \mathbb{R} \rightarrow [0, \infty)$ be a continuously differentiable function. Prove that

$$\left| \int_0^1 f(x)^3 dx - f(0)^2 \int_0^1 f(x) dx \right| \leq \max_{0 \leq x \leq 1} |f'(x)| \left(\int_0^1 f(x) dx \right)^2.$$

299. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a twice continuously differentiable function such that

$$|f''(x) + 2xf'(x) + (x^2 + 1)f(x)| \leq 1$$

for all $x > 0$. Prove that $\lim_{x \rightarrow \infty} f(x) = 0$.

300. For any given $x \in (0, \pi/2)$ determine which of $\tan(\sin(x))$ and $\sin(\tan(x))$ is bigger.

301. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function satisfying $f(0) = 1$ and $f'(0) = 0$, and further

$$f''(x) - 5f'(x) + 6f(x) \geq 0 \quad \text{for all } x \geq 0.$$

Show that for all $x \geq 0$, it holds that

$$f(x) \geq 3e^{2x} - 2e^{3x}.$$

302. Let $0 < a < b$. Prove that

$$\int_a^b (x^2 + 1)e^{-x^2} dx \geq e^{-a^2} - e^{-b^2}.$$

303. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function with the property that

$$f'(t) > f(f(t)) \quad \text{for all } t \in \mathbb{R}.$$

Show that

$$f(f(f(t))) \leq 0 \quad \text{for all } t \geq 0.$$

304. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function with the property that $f(0) = 0$. Show that there exists some $c \in (-\pi/2, \pi/2)$ such that

$$f''(c) = f(c)(1 + 2 \tan(c)^2).$$

305. (*) Define $f(x) = \sin(x)/x$. Show that for all $x > 0$ and $n \geq 1$, it holds that

$$|f^{(n)}(x)| < \frac{1}{n+1},$$

where $f^{(n)}$ denotes the n th derivative of f .

306. Consider the set of continuous functions $f : [0, 1] \rightarrow \mathbb{R}$ satisfying

$$f(x) + f(y) \geq |x - y| \quad \text{for all } x, y \in [0, 1].$$

Find the minimal value of

$$\int_0^1 f(x) dx$$

for functions in this set.

307. Consider a differentiable function $f : \mathbb{R} \rightarrow (0, \infty)$ with the property that for some constant $L > 0$, it holds that

$$|f'(x) - f'(y)| \leq L|x - y| \quad \text{for all } x, y \in \mathbb{R}.$$

Show that for all $x \in \mathbb{R}$, it holds that

$$f'(x)^2 < 2Lf(x).$$

308. Find all differentiable functions $f : (0, \infty) \rightarrow \mathbb{R}$ with the property that

$$f(b) - f(a) = (b - a)f'(\sqrt{ab}) \quad \text{for all } a, b > 0.$$

309. Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a twice differentiable function such that $2f'(x) + xf''(x) \geq 1$ for all $x \in (-1, 1)$. Prove that

$$\int_{-1}^1 xf(x) dx \geq \frac{1}{3}.$$

310. Find all twice continuously differentiable functions $f : \mathbb{R} \rightarrow (0, \infty)$ such that

$$f''(x)f(x) \geq 2f'(x)^2 \quad \text{for all } x \in \mathbb{R}.$$

311. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that have a continuous second derivative and for which the equality

$$f(7x + 1) = 49f(x)$$

holds for all $x \in \mathbb{R}$.

312. If f has continuous derivative on $[-1, 1]$, find

$$\lim_{h \rightarrow 0^+} \int_{-1}^1 \frac{hf(x)}{h^2 + x^2} dx,$$

assuming the limit exists.

313. Let $f : [0, 1] \rightarrow \mathbb{R}$ be twice differentiable on $[0, 1]$ with $f''(x) > 0$ on $[0, 1]$. Show that

$$4 \int_{1/2}^1 f(x) dx \leq f(1) + \int_0^1 f(x) dx.$$

314. If f has continuous third derivative on $[a, b]$ and has at least three roots in $[a, b]$, show that

$$\max_{x \in [a, b]} |f(x)| \leq \frac{(b-a)^3}{6} \max_{x \in [a, b]} |f^{(3)}(x)|.$$